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ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE)
PHYSICS EXAMINATION 2009

BOOKLET CODE: 00

NUMBER OF ITEMS:

SUBJECT CODE: 00

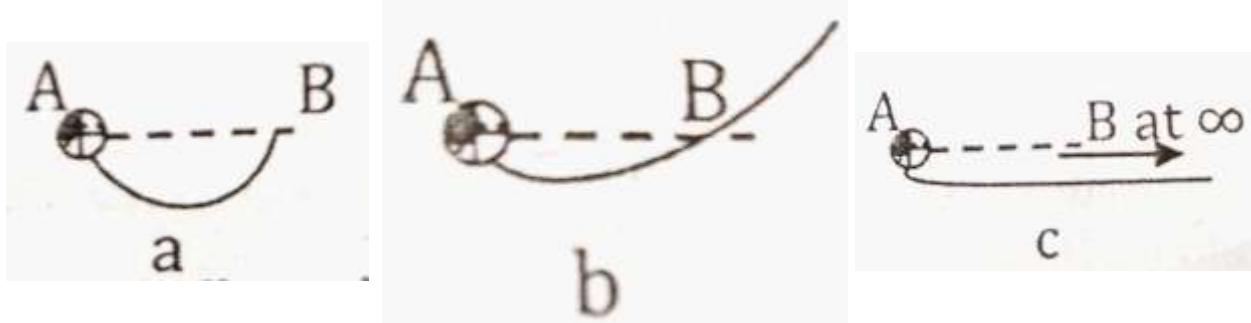
TIME ALLOWED: 00

Constant	Symbol	Value
Acceleration due to gravity	g	10m/s^2
Permittivity of vacuum	ϵ_0	$8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$
Charge of an electron	e	$1.6 \times 10^{-19} \text{ coulomb}$
Universal gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}$
Density of water	ρ	1000kg/m^3
Planck's constant	h	$6.63 \times 10^{-34} \text{ J.s}$
Electron mass		$9.11 \times 10^{-31} \text{ kg}$
Elementary (unit) charge		$1.602 \times 10^{-19} \text{ C}$
$\ln 2 = 0.693$		
$\sin 30^\circ = \cos 60^\circ = 0.5$		
$\sin 37^\circ = \cos 53^\circ = 0.6$		
$\sin 45^\circ = \cos 45^\circ = 0.707$		
$\sin 53^\circ = \cos 37^\circ = 0.8$		
$\sin 60^\circ = \cos 30^\circ = 0.866$		

1. The range of cannon-ball fired horizontally from a laboratory table is equal to $8/3$ times the height of the table. What is the direction of velocity vector of the projectile as it strikes the ground?

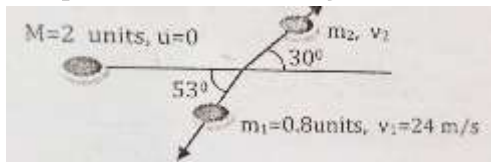
- ☐ A. 45° from the horizontal
- ☐ B. 53° from the horizontal
- ☐ C. 60° from the horizontal
- ☐ D. 37° from the horizontal

2. Galileo carried out a **thought experiment** on motion. The line between A and B represents a surface and the center of the ball is placed at A. what would happen in each of the three situations shown in the figure below, without friction when the ball is released?



- ☐ A. The ball would never reach point B in all instances.
☐ B. The ball would reach B in all instances even if point B keeps on receding to infinity in (c).
☐ C. The ball would reach point B in (a) and (b) and not in (c).
☐ D. The ball would reach point B only in (a).

3. A uranium atom initially at rest has a mass of 2 units and spontaneously disintegrated into two fragments. One of the fragments has a mass of 0.8 units and moves at a speed of 24m/s as shown in figure below. What is the speed of the second fragments?



- ☐ A. 25.6m/s
☐ B. 21.3m/s
☐ C. 16m/s
☐ D. 14.8m/s

4. A 1kg object explodes and breaks up into three pieces. One piece has a mass of 400g and has a velocity of $(15\mathbf{i} + 10\mathbf{j})\text{m/s}$. a second piece has a mass of 250g and has a velocity of $(25\mathbf{i} - 10\mathbf{j})\text{m/s}$. What is the velocity of the third piece?

- ☐ A. $(-4.3\mathbf{i} + 35\mathbf{j})\text{m/s}$
☐ B. $(-35\mathbf{i} - 4.3\mathbf{j})\text{m/s}$
☐ C. $(35\mathbf{i} - 4.3\mathbf{j})\text{m/s}$
☐ D. $(-35\mathbf{i} - 4.3\mathbf{j})\text{m/s}$

5. Which of the following statements is not correct?

- ☐ A. Dynamic equilibrium is an equilibrium in which a body is moving at steady velocity without net force or torque acting on it.
☐ B. Static equilibrium is an equilibrium in which a body is at rest without net force or torque acting on it.
☐ C. Couples are set of forces with a resultant torque but no net force.
☐ D. The necessary and sufficient condition for equilibrium of a rigid body is to have the sum of all torques acting on it is zero.

6. A spring with a spring constant of 300N/m is stretched 20cm in 2 seconds. What is the power applied to stretch the spring?

- ☐ A) 3W
- ☐ B) 6W
- ☐ C) 30KW
- ☐ D) 60KW

7. A 50KW machine is used to lift heavy objects to the top of a building under construction. If this machine is replace with a new machine of 100KW power, then which of the following statements is not true about the new machine?

- ☐ A. The new machine can raise objects with the same mass at twice the speed of the old machine.
- ☐ B. The new machine can raise objects with twice as much mass at the same speed as the old machine.
- ☐ C. The new machine takes twice as much time to do the same work as the old machine.
- ☐ D. The new machine can do twice as much work in the same time as the old machine.

8. Which of the following is correct about a work done for dissipative force?

- ☐ A. Path dependent
- ☐ B. The same as the loss in kinetic energy.
- ☐ C. The same as the gain in potential energy.
- ☐ D. The same as the potential energy stored in spring.

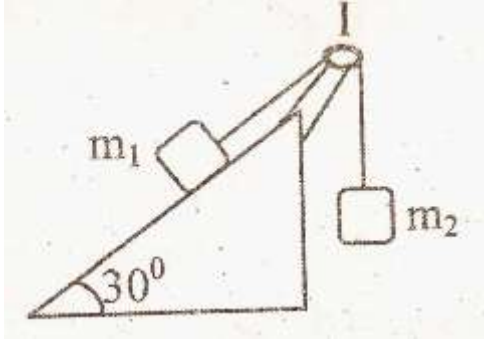
9. A disk of radius R with uniform distribution of mass M rotates about an axis perpendicular to its plane at the rim with angular speed ω the moment of inertia of the disc about the axis through the center is $\frac{1}{2}MR^2$. What is the kinetic energy of the disk?

- ☐ A. $\frac{1}{4}MR^2\omega^2$
- ☐ B. $\frac{1}{3}MR^2\omega^2$
- ☐ C. $\frac{1}{2}MR^2\omega^2$
- ☐ D. $\frac{4}{4}MR^2\omega^2$

10. A pulley is rotating at a frequency of 20rev/min. a motor speeds up the wheel so that 30s later the frequency of the motion is 60rev/min. How far a point on the pool 0.30m from its center will has travelled during that period?

- ☐ A) 125.7m
- ☐ B) 0.13m
- ☐ C) 37.68m
- ☐ D) 4.2m

11. A block of mass $m_1=10\text{kg}$ placed on a frictionless incline plane connected to mass $m_2=15\text{kg}$ using a string of negligible mass passing over a pulley of moment of inertia $I=0.25\text{kgm}^2$ and radius $R=10\text{cm}$ as shown in the diagram. What is the torque on the pulley?

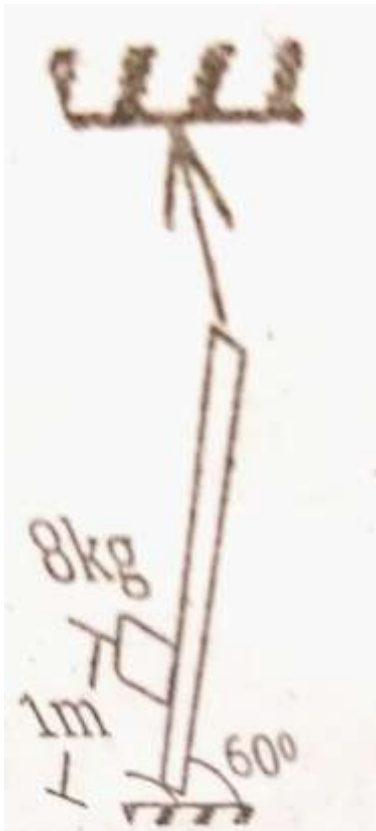


- ☐ A. 2.5Nm
- ☐ B. 4Nm
- ☐ C. 5Nm
- ☐ D. 12Nm

12. Which of the following statements is correct about the first condition of equilibrium?

- ☐ A. The net force on an object must be zero.
- ☐ B. The sum of all torques must be zero.
- ☐ C. Clockwise moments should balance counter clockwise moments.
- ☐ D. The torque produced by a force should be perpendicular to the force.

13. A 4m long uniform board of mass 10kg hinged on the ground at one end is hanging by a string at the other end. A block of mass 8kg is placed at 1m from the hinge on the ground as shown in the diagram below. What is the tension in the string?



- ☐ A. 10N
- ☐ B. 50N
- ☐ C. 70N
- ☐ D. 110N

14. Which of the following refers to the definition of strain as a ratio?

- ☐ A. Extension to the original length.
- ☐ B. Force applied to area of the object normal to the force.
- ☐ C. Stress to Young modulus.
- ☐ D. Tensile to applied force.

15. Which of the following is the application of Pascal's law?

- ☐ A. Hydraulic brake in vehicles.
- ☐ B. Siphon to remove fluids.
- ☐ C. Working of submarines.
- ☐ D. Sucking of liquid into a syringe.

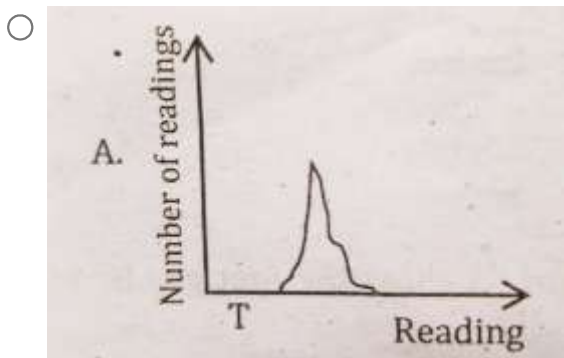
16. A large (very large in height and cross-section) water open at the top has a small hole through its side near the bottom. To what height should the water be filled, to double the speed of water that flow out of the hole when it was filled to a height h ?

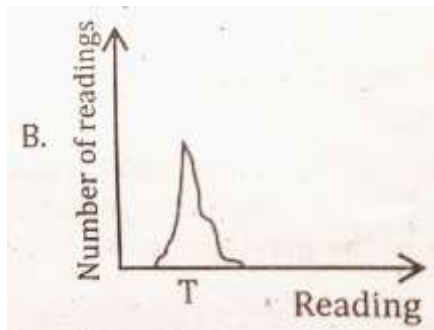
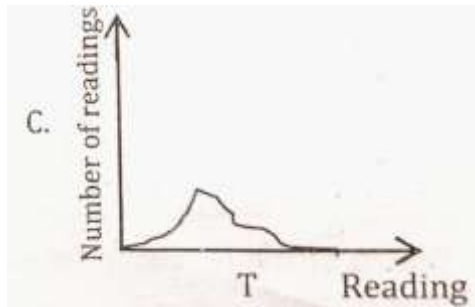
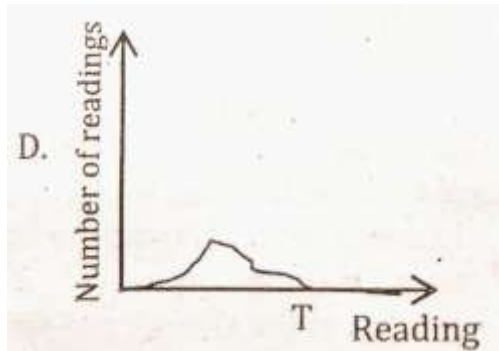
- ☐ A) h
- ☐ B) $2h$
- ☐ C) $3h$
- ☐ D) $4h$

17. If a precision of surveyor measured a distance of 500 meters /between two points in a village/ using a meter that has centimetre division on it, then what is the measurement result?

- ☐ A) 500m
- ☐ B) 500.0m
- ☐ C) 500.00m
- ☐ D) 500.000m

18. From the figures given below, which one of them shows precise but not accurate measurements? T represents the true value.



☐☐☐

19. If vector $\vec{P}$ is 6m in the x-direction and vector $\vec{q}$ is 8m in the y-direction. Then what is the value of the vector $\vec{p} + \vec{q}$?

- ☐ A. 10m at 53° from x-direction.
- ☐ B. $\sqrt{136}m$ at 45° from x-direction.
- ☐ C. 10m at 37° from x-direction.
- ☐ D. $\sqrt{136}m$ at 30° from x-direction.

20. Given vectors $\vec{A} = \sqrt{3}\vec{i} + \vec{j}$ and $\vec{B} = \sqrt{3}\vec{i} - \vec{j}$. What is the smaller angle between the vectors?

- ☐ A) 60°
- ☐ B) 53°
- ☐ C) 45°
- ☐ D) 30°

21. If $\vec{A} = \vec{x}\frac{m}{s} + 3\frac{m}{s}\vec{z}$ and $\vec{B} = 2\frac{m}{s}\vec{y} + 3\frac{m}{s}\vec{z}$, then what is the value of $|\vec{A} \times \vec{B}|$?

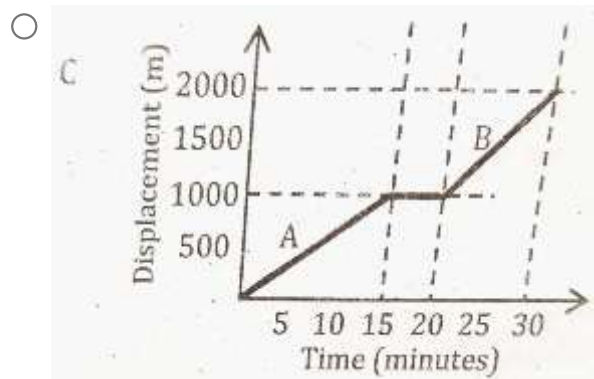
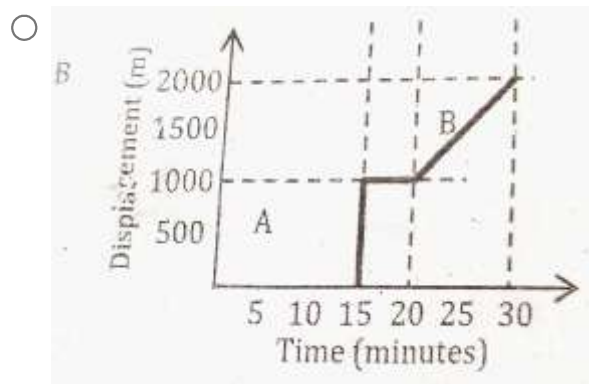
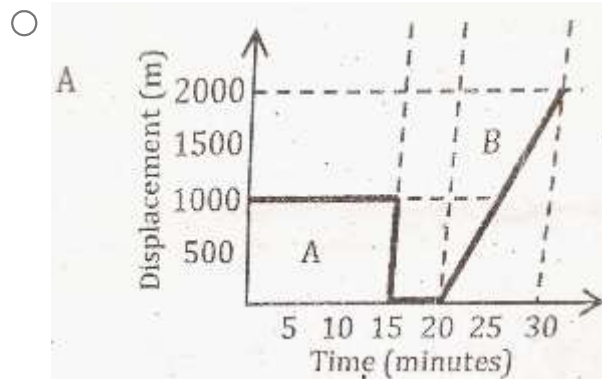
- ☐ A) $\sqrt{10}\frac{m}{s}$

- ☐ B) $\sqrt{13} \frac{m}{s}$
- ☐ C) $7 \frac{m}{s}$
- ☐ D) $\sqrt{130} \frac{m}{s}$

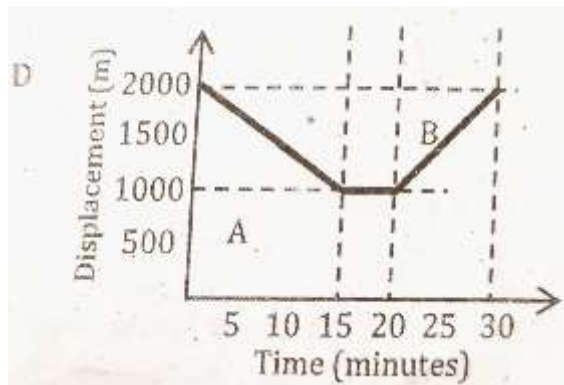
22. A cricketer throws a ball vertically upwards with initial speed of 20m/s. How long is it in the air before it returns to the cricketer's hands?

- ☐ A) 2s
- ☐ B) 10s
- ☐ C) 1.5s
- ☐ D) 4s

23. Abeba walks to school. She walks 1km in 15minutes. She meets her friend Makeda they talk for 5 minutes and then carry on walking to school. They walk 800m in 10minutes. Which of the following displacement time graph shows Abeba's journey to school?



○



24. A fun making boy swings a blinking ball tied to the end of a string in a vertical circle. If the radius of the circle is 0.5m, then what should be the minimum speed of the ball at the bottom of the circle to make a complete circular motion at the top?

- A) 5m/s
 ○ B) $\sqrt{5}$ m/s
 ○ C) $\sqrt{15}$ m/s
 ○ D) $2\sqrt{5}$ m/s

25. The activity of radioactive substance reduces from 2000Bq to 1000Bq in 10^6 seconds. What is the decay constant?

- A. 6.93×10^5 , decay/s
 ○ B. 6.93×10^{-7} , decay/s
 ○ C. 2.00×10^6 , decay/s
 ○ D. 5.00×10^{-7} , decay/s

26. Which of the following phenomena clearly demonstrates the particle nature of light?

- A. Refraction of light by prism.
 ○ B. The photoelectric effect.
 ○ C. Diffraction of light by double slit.
 ○ D. Dispersion of white light.

27. Which of the following statements is correct?

- A. Irreversible process is a process that can be reversed through finite change in the system.
 ○ B. Complete conversion of energy from a hot source into work is possible.
 ○ C. There are processes in which there is an overall decrease in entropy.
 ○ D. Entropy is a measure of the amount of disorder.

28. A certain heat engine draws 500cal/s from a water bath at 27°C and rejects 400 calories to a reservoir at a lower temperature. What is the efficiency of this engine?

- A) 80%
 ○ B) 75%

- ☐ C) 25%
- ☐ D) 20%

29. Three metals are in contact and in thermal equilibrium. The metals have different specific heat capacity. What can be concluded about the temperature of the metals?

- ☐ A. Since they have different specific heat capacity, they must have different temperature at thermal equilibrium.
- ☐ B. They must have the same temperature at thermal equilibrium.
- ☐ C. They must have different internal energy.
- ☐ D. They must have different kinetic energy.

30. Metal A has a coefficient of linear expansion that is three times the coefficient of linear expansion of metal B. How does their coefficient of areal expansion, Y , compare?

- ☐ A) $Y_A = 2Y_B$
- ☐ B) $Y_A = 3Y_B$
- ☐ C) $Y_A = 4Y_B$
- ☐ D) $Y_A = 6Y_B$

31. On an unknown scale x , the ice point temperature is $50^\circ x$ and the steam-point temperature is $200^\circ x$. If the temperature is $60^\circ C$, then what is the reading in $^\circ x$?

- ☐ A) $90^\circ x$
- ☐ B) $140^\circ x$
- ☐ C) $15^\circ x$
- ☐ D) $333^\circ x$

32. A block of mass 5kg hanging at the end of spring scale is totally submerged in a beaker of water. The beaker is large enough to avoid overflow of water. If the density of the block is $2.5 \times 10^3 \text{ kg/m}^3$, then what is the reading of the scale?

- ☐ A) 3kg
- ☐ B) 2kg
- ☐ C) 1kg
- ☐ D) 0kg

33. An electric generator has a coil of 400 turns and cross-sectional area 5 m^2 rotating at angular frequency 0.524 rad/s in a uniform magnetic field of 0.2 T , what is the induced e.m.f produced from the electric generator in each second?

- ☐ A) 209.6V
- ☐ B) 104.8V
- ☐ C) 200V
- ☐ D) 182.4V

34. The primary coil of a transformer is connected to a battery, a resistor, and a switch. The secondary coil is connected to an ammeter. What does the ammeter show when the switch is thrown closed?



- ☐ A. Zero current as transformer is not working on DC.
- ☐ B. A nonzero current for a short instant.
- ☐ C. A steady current.
- ☐ D. emf will be induced but not current.

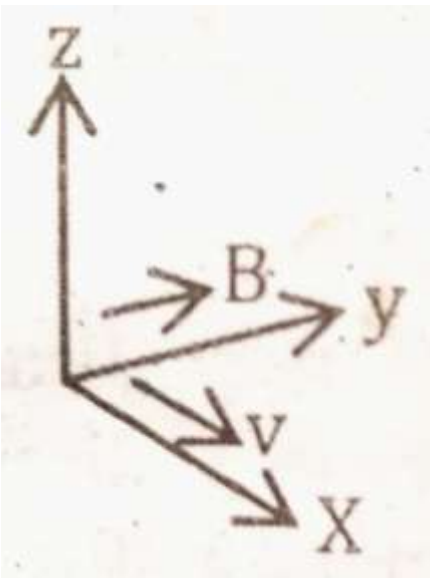
35. Which of the following statement is correct?

- ☐ A. The direction of induced e.m.f is in such a direction as to reinforce the change produces it.
- ☐ B. Self-inductance of a coil is e.m.f divided by a magnitude of rate change of current with time.
- ☐ C. The induced e.m.f in a coil is inversely proportional to its number of turns.
- ☐ D. Self-inductance of a coil is inversely proportional to the cross sectional area of the coil.

36. The induced electromotive force in a current carrying coil placed in a magnetic field does not depend on which of the following?

- ☐ A. The speed at which the coil moves if the relative motion does not result in changing the magnetic flux.
- ☐ B. The change of magnetic flux density.
- ☐ C. The length of the coil that is moving in a direction perpendicular to that of the magnetic field.
- ☐ D. The area of the coil outside the field.

37. A positively charged particle of charge $q=3.2 \times 10^{-19}\text{C}$ is moving parallel to x-axis with speed 1000m/s in a magnetic field of 0.5T parallel to y-axis as shown in the diagram below. What is the magnetic force on the charge?



- ☐ A. $8.0 \times 10^{-16}\text{N}$ in +ve Z
- ☐ B. $3.2 \times 10^{-16}\text{N}$ in +ve x
- ☐ C. $8.0 \times 10^{-17}\text{N}$ in +ve x

- ☐ D. $1.6 \times 10^{-16} \text{N}$ in + ve Z

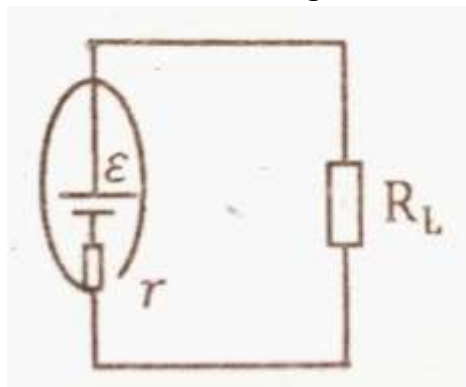
38. A charged particle moves perpendicular to a uniform magnetic field. Which statement is correct about the motion of the particle?

- ☐ A. Both kinetic energy and linear momentum of the particle change.
- ☐ B. The kinetic energy of the particle change but not its linear momentum.
- ☐ C. The linear momentum of the particle changes but not its kinetic energy.
- ☐ D. Both the energy and momentum do not change.

39. Which of the following is not correct about magnetic field?

- ☐ A. Magnetic field lines are closed.
- ☐ B. Magnetic monopoles exist.
- ☐ C. The closer the field lines are, the stronger the magnetic field.
- ☐ D. Magnetic field is produced by moving charge.

40. Which of the following is not correct about the circuit shown below?



- ☐ A. Voltage across R_L is $\frac{R_L}{r+R_L} \mathcal{E}$
- ☐ B. Current flowing through R_L is $\frac{\mathcal{E}}{r+R_L}$
- ☐ C. Current flowing through r is $\frac{r}{r+R_L} \mathcal{E}$
- ☐ D. Power drop in R_L is $\frac{R_L}{(r+R_L)} \mathcal{E}^2$

41. A galvanometer of full-scale deflection 10mA is to be converted into a 0 $\rightarrow$ 10.0A ammeter. If its coil has a resistance of 100Ω , then what value of shunt resistance must be fitted?

- ☐ A) 0.1Ω
- ☐ B) 10Ω
- ☐ C) 9.99Ω
- ☐ D) 0.999Ω

42. If a cell supplies a current of 0.9A when connected to a 2Ω resistor and a current of 0.3A , then what will be the internal resistance of the cell when connected to a 7Ω resistor?

- ☐ A) 2Ω

- ☐ B) 1.2Ω
- ☐ C) 1Ω
- ☐ D) 0.5Ω

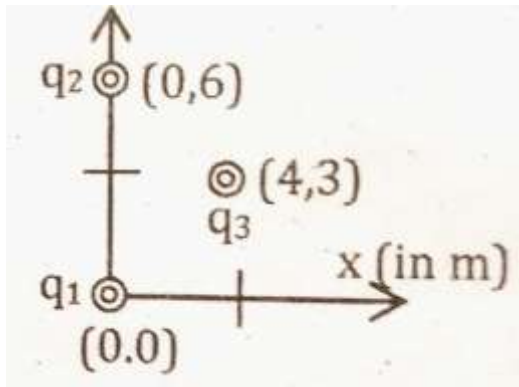
43. The capacitance of a parallel plate capacitor is 10. If the distance between the plates is halved, keeping all other quantities constant, then what is the new capacitance of the capacitor?

- ☐ A) $0.2\mu\text{F}$
- ☐ B) $5\mu\text{F}$
- ☐ C) $10\mu\text{F}$
- ☐ D) $20\mu\text{F}$

44. Two positive charges of magnitude $2.4 \times 10^{-9}\text{C}$ and $1.6 \times 10^{-9}\text{C}$ are separated in air by a distance of 2.0cm. How much electrostatic potential is produced by the two charges at midway between them?

- ☐ A) 3.6KV
- ☐ B) 0.72 KV
- ☐ C) 2.16 KV
- ☐ D) 1.44 KV

45. A charge of $q_1 = -10.0 \times 10^{-9}\text{C}$, $q_2 = +10.0 \times 10^{-9}\text{C}$ and $q_3 = -2.5 \times 10^{-8}\text{C}$ are placed on x-y plane as shown in the figure below. What is the electric force on q_3 by q_1 and q_2 ?



- ☐ A. $1.08 \times 10^{-7}\text{N}$ in + ve y direction
- ☐ B. $9.00 \times 10^{-7}\text{N}$ in - ve y direction
- ☐ C. $1.42 \times 10^{-7}\text{N}$ in + ve x direction
- ☐ D. $1.42 \times 10^{-7}\text{N}$ in - ve x direction

46. In a Young's experiment, there are two slits, rather than one. What do you think the reason?

- ☐ A. To increase the intensity.
- ☐ B. One slit is for frequency the other for wavelength.
- ☐ C. To create a path length difference.
- ☐ D. One slit is for electric fields and the other is for magnetic fields.

47. A light of wavelength 590nm is incident normally on a grating with spacing $1.67 \times 10^{-6}\text{m}$. How many orders of diffraction maxima can be obtained (including the zero order)?

- ☐ A) 1

- ☐ B) 2
- ☐ C) 3
- ☐ D) 4

48. Which of the following statements is correct?

- ☐ A. Interference is not superposition of waves.
- ☐ B. Constructive interference results in decrease in the intensity.
- ☐ C. Destructive interference results in an increases in the intensity.
- ☐ D. Path difference which is integer multiple of wavelength results in constructive interference.

49. Which of the following is not correct about a simple harmonic oscillator?

- ☐ A. Total mechanical energy is conserved.
- ☐ B. The acceleration is directly proportional to and opposite in direction to displacement
- ☐ C. The period of small oscillation of a pendulum with string length l , a bob mass m attached to its end is $T = 2\pi \sqrt{\frac{l}{g}}$.
- ☐ D. The frequency of an oscillating mass attached to a massless spring is independence of the spring constant.

50. Which one of the following indicates the wavelength of a travelling wave?

- ☐ A. The distance between an adjacent trough and a peak.
- ☐ B. The amplitude of the wave.
- ☐ C. The distance between a node and antinodes.
- ☐ D. The distance between two adjacent points which are in phase.

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